

# Joe Badaczewski

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## Senior Software Development Engineer with 14+ years of proven experience building large-scale, full-stack systems and mobile applications using C#, Node.js, React Native, and TypeScript

### Core skills

C#, Node.js, TypeScript, React, React Native, modern front-end architecture, front-end development and module bundling, micro-frontend systems at scale, data-heavy UI and data visualization tooling, 3D visualizations with three.js, driving cross-team standards and best practices for web development, a well-established technical design and documentation habit, full-stack system design, platform SDK and API design for multi-team consumption, composable guardrails and responsible AI (PII handling, audit logging), event-driven architectures (AWS EventBridge, SQS), streaming infrastructure (SSE, WebSockets), multi-tenant platform infrastructure, AWS (EKS, Fargate, DynamoDB, RDS, S3, EventBridge), PostgreSQL and Redis, observability (metrics, tracing, alerting, Grafana), implementing OIDC providers with fine-grained RBAC, large-scale project management, cross-team collaboration and customer partnership, maintaining operational excellence in distributed systems with Kubernetes, Agile software development and continuous delivery of value, mentorship and technical leadership, working with high ownership and autonomy, working well under pressure, optimizing user experiences for performance, accessibility, usability, and scalability.

### Professional Experience

**Softwriters, Inc.**, Pittsburgh, PA  
*Senior Software Engineer*

03/2024 - Present

- Lead architect on our shared authorization layer that uses JWKS and Keycloak to federate access to our multi-tenant cloud infrastructure; deployed using Kubernetes on an AWS EKS instance.
- Senior lead of FrameworkErX, a platform written in TypeScript that uses AI capabilities to parse and automate medication order entry. Focused on responsible AI usage with proper guardrails for handling PII and comprehensive audit logging.
- Senior lead of FrameworkCourier, an application written in C# that is responsible for tracking the chain of custody of prescriptions through a system of pharmacies and facilities.
- Designed and implemented a suite of mobile applications written in React Native that make pharmacies more efficient in packaging and reviewing prescriptions.
- Planned and developed a cross-team project that used Large Language Models to give pharmacies better auditing tools by translating natural language to executable SQL statements.
- Built a proof of concept written in full-stack F# that uses HTTP-only cookie authorization with an OIDC provider (Keycloak). This prototype enables retail pharmacies to independently build on top of FrameworkLTC's infrastructure.
- Drive cross-team standards and best practices for web development, architecting a shared React and TypeScript front-end ecosystem and design documentation adopted by multiple product teams.
- Design and ship mission-critical UI systems for complex, data-heavy pharmacy workflows, collaborating directly with internal customers and stakeholders to align tooling with real-world operations while driving performance, scalability, and maintainability.
- This role requires an expert skill level in C#, TypeScript, React, React Native, modern front-end architecture, Git, cloud services, service-oriented architecture at scale, Agile software development, and interfacing with multiple team leaders to achieve common product goals.

**Amazon Web Services**, Seattle, WA (remote)  
*Front-end Engineer II*

05/2022 - 03/2024

- Led the design, development, and delivery of CloudWatch Logs Anomaly Detection: a tool for the CloudWatch Logs team that enabled customers to detect elevated errors and patterns and decreased time-to-action for customer-impacting events.

- Spearheaded the migration of CloudWatch Logs Insights from a legacy backend service to a modern service with more granular role-based authorization without affecting uptime. The new authorization strategy provided better security for 2M+ requests per day.
- Managed the onboarding process of three new developers into a rigorous on-call environment for a 99.9% availability service with a 15-minute response time.
- Volunteered for Code.org's Hour of Code and led the instruction of 3 one-hour sessions to elementary and high-school students on the importance of STEM careers and basic programming concepts.
- This role required an expert skill level in leading large-scale projects with many stakeholders, front-end system design, React, TypeScript, working well under pressure, mentorship, and managing optimal accessibility and performance for a complex system of micro-frontends.

**Homee, Inc.**, Orlando, FL (remote)

*Senior Software Engineer, Team Lead*

10/2021 - 05/2022

- Hired and led a team of mobile engineers responsible for four industry-grade mobile applications for iOS/Android that simplified the assessment and estimation of insurance claims.
- Managed an IAM service using Keycloak (an open-source, OIDC-compliant server) that served 1000+ active users.
- Led the development of complex features such as turn-by-turn navigation, location-based consumer and provider matching, offline syncing and fault tolerance, optical character recognition for analyzing estimates and receipts, and an intuitive scheduling interface including efficient native gestures and animations.
- This role required an expert skill level in federated GraphQL, Node.js/Express/Next.js, React, React Native, Redux, TypeScript, AWS, Postgres, Redis, and OIDC.

**Softwriters, Inc.**, Pittsburgh, PA

*Senior Software Engineer*

05/2020 - 10/2021

- Team lead for a suite of two iOS and Android applications that streamlined the packaging, reviewing, and delivery of controlled and uncontrolled substances from long-term care pharmacies to LTC facilities.
- Lead architect in designing and deploying a distributed system responsible for tracking the chain of custody and courier history of millions of prescriptions.
- This role required an expert skill level in C#, TypeScript, cloud services, large-scale system design, and service-oriented architecture.

**Proofpoint**, Pittsburgh, PA

*Senior Software Engineer*

09/2019 - 05/2020

- Lead developer and code owner of multiple enterprise React/Redux/TypeScript applications that provide training assignments and exercises to 15,000+ concurrent users.
- Organized architectural direction and philosophy for a team of front-end and back-end developers.
- This role required an expert skill level in React, Node.js (NestJS), TypeScript, CI/CD, Git, and the principles behind scaling a multi-contributor codebase within an Agile environment.

**Idelic**, Pittsburgh, PA

*Software Engineer*

02/2019 - 09/2019

- Lead developer for web applications aiding terminal managers of trucking companies with monitoring and improving driver performance and safety.
- Utilized microservices via Docker to create a multi-tiered platform that achieved high levels of concurrency, availability, and reliability.
- This role required an expert skill level in Node.js, React, React Native, Redux, and Docker. Also required knowledge of Scala, PostgreSQL, and test-driven development.

**PSI Services, LLC**, Pittsburgh, PA

*Software Engineer*

10/2017 - 02/2019

- Software engineer responsible for creating and maintaining HR solution software that assists Fortune 500 companies with selecting the best available candidates.
- Interfaced with Industrial and Organizational psychologists to develop algorithms that assess a candidate's potential for success.
- This role required extensive knowledge in C#, unit testing, OOP, Angular, JavaScript ES6, cloud computing, and CI/CD.

**Brunner**, Pittsburgh, PA

*Senior Visual Developer*

09/2014 - 10/2017

- Senior developer responsible for planning and building web applications for a myriad of companies in the financial and e-commerce/retail industries.
- Led a team of front-end and back-end developers in delivering highly performant and low-latency web and mobile applications that drove leads for our customers and managed billions of dollars in investments.
- This role required expert-level knowledge of C#, TypeScript, and project management.

### **Education**

**M.S. Multimedia Technology**, 2012 - 2014

Duquesne University, Pittsburgh, PA

**B.A. in Digital Media Arts**, 2008 - 2012

Duquesne University, Pittsburgh, PA